

KARACHI SCHOOL OF BUSINESS & LEADERSHIP (KSBL)

Indus Hospital Operations Intelligence Database

Project Proposal & Implementation Plan

A highly optimized, professional relational database design mapping the core patient-care pathways and donor-funding mechanisms. Tailored to model real-world hospital operational dynamics under supervisor guidance.

PREPARED BY

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ACADEMIC PROGRAM

BS Computer Science - Data Science

PROJECT DOMAIN

Healthcare Data Engineering & Business Intelligence

INTERNSHIP DURATION

July 13 – July 24, 2026

1. INTRODUCTION & PROJECT VISION

Standard clinical databases focus primarily on localized, transactional entries—registering patients, producing bills, and scheduling basic visits. While essential for daily administration, they often lack the schema mapping required for deep, strategic resource planning and operational intelligence.

The **Indus Hospital Operations Intelligence Database** is designed as a high-impact, centralized operational intelligence model. Built around the unique operational landscape of **Indus Hospital & Health Network** (Korangi Campus), the project models a zero-fee, donation-funded clinical architecture where patient care flows are directly integrated with funding sustainability.

Strategic Modeling Choice: Philanthropy over Fee-for-Service

A billing/finance-ledger table was deliberately excluded from this database model. Unlike standard fee-for-service hospital designs, Indus Hospital operates on a strict zero-fee basis where patient care is never billed, invoiced, or co-pay. Donations are modeled as pure direct operational inflows, accurately reflecting the hospital's authentic philanthropic care model.

2. PATIENT-CARE PATHWAYS (SUPERVISOR-GUIDED EXPANSION)

Following direct guidance and discussions with the internship supervisor, the project scope was expanded to comprehensively capture the hospital's actual patient-care pathways. Instead of tracking inpatient admissions in isolation, the database structure now models how different patients move through and consume resources across the entire hospital system:

Together, **OPD (Outpatient) Visits**, **Emergency Visits**, **Admissions (Inpatient Care)**, and **Surgeries / Procedures** represent the four primary care pathways. This structure allows us to measure resource constraints and throughput metrics realistically.

3. MAIN PROJECT DELIVERABLES

- **Database Schema:** A clean, fully normalized 3NF (Third Normal Form) relational database design.
- **Relational Synthetic Dataset:** Programmatically generated records (using Python's pandas and Faker) containing ~77,000+ logically linked rows preserving strict referential integrity.
- **Advanced SQL Analytics Engine:** Production-ready scripts containing optimized analytical queries (Window functions, CTEs, self-joins) for operational key performance indicators.
- **Executive Dashboard:** An interactive Tableau Business Intelligence dashboard modeling capacity, workflow efficiency, and donor contribution matches.

4. COMPREHENSIVE RELATIONAL SCHEMA

The revised database framework comprises exactly 11 tables (including reference and junction tables), balancing structural detail with relational integrity:

Table Name	Primary & Foreign Keys	Operational & Business Intelligence Purpose
1. Patients	PatientID (PK)	Primary demographics (Age, Gender, City, Reg Date) and verified national CNIC formatting.
2. Diseases	DiseaseID (PK)	Standardized clinical reference table mapping major disease classifications.
3. PatientDiseases	PatientID (FK), DiseaseID (FK)	Junction table to resolve comorbidity states (a patient with diabetes and chronic kidney disease).
4. Doctors	DoctorID (PK)	Details clinical staff, departmental specialty alignment, experience, and shifts.
5. Wards	WardID (PK)	Defines specific hospital departments/units (e.g., ICU, Pediatric Ward, HDU, General Medical).
6. Beds	BedID (PK), WardID (FK)	Maps physical beds to their assigned wards to track real-time and historical capacity constraints.
7. Admissions (Inpatient)	AdmissionID (PK), PatientID (FK), BedID (FK), DoctorID (FK)	Core inpatient transaction table tracking lengths of stay, readmissions, and discharges.
8. OPD Visits	OPD_ID (PK), PatientID (FK), DoctorID (FK)	Tracks same-day outpatient interactions, diagnostic conclusions, and follow-up flags.
9. Emergency Visits	EmergencyID (PK), PatientID (FK)	Models ER volume, triage status (Critical, Urgent, Non-urgent), and outcome pathways.
10. Surgeries / Procedures	SurgeryID (PK), PatientID (FK), DoctorID (FK)	Tracks operative procedures, assigned Operating Theatres, and cancellation reasons.
11. Donations	DonationID (PK)	Captures incoming philanthropic funding channels (Zakat, Sadaqah, Endowment, General Fund).

Updated Data Volumes

- **Patients:** 12,000 unique individuals
- **Diseases:** 40 standardized profiles | **Patient-Disease Bridge:** ~18,000 links
- **Doctors:** 180 specialist physicians | **Wards:** 20 units | **Beds:** 500 bed capacity
- **Admissions (Inpatients):** 18,000 historical rows

- **OPD Visits:** 15,000 records | **Emergency Visits:** 6,000 entries | **Surgeries:** 4,000 records
- **Donations:** 4,000 donation entries
- **Total Dataset Complexity:** 11 physical tables representing over 77,000+ rows.

5. IMPLEMENTATION PLAN (JULY 13 – 24, 2026)

To successfully deliver a clean, robust, and complete database ecosystem, the 10-day timeline is segmented into disciplined sprints:

Timeline Block	Operational Focus	Deliverables & Scope Milestones
Days 1–2 (July 13–14)	Schema Design & ERD Setup	Determine complete database structure, map entity constraints, draft the 11-table ERD, and secure supervisor approval.
Days 3–4 (July 15–16)	Python Data Generation	Develop clean Python generator scripts (Faker/pandas). Secure chronological constraints across the four care pathways.
Days 5–6 (July 17–20)	Database Setup & Population	Draft standard schema creation scripts, set referential integrity indexes, import simulated data, and test foreign key constraints.
Days 7–8 (July 21–22)	Complex Query Formulation	Write analytical queries (CTEs, Window Functions) addressing patient throughput, surgery cancellations, and triage outcome distributions.
Days 9–10 (July 23–24)	BI Dashboard & Documentation	Connect database views to Tableau, build the executive-level operations board, compile documentation, and deliver the project wrap-up.

6. KEY ANALYTICAL QUESTIONS RESOLVED

By consolidating clinical operations and donor funding into a relational format, the database enables executives to immediately run high-value business queries:

- **Patient-Care Pathways Distribution:** What is the exact volume, resource consumption, and demographic distribution of patients between OPD, Emergency, and Inpatient pathways?
- **Surgical Cancellation Efficiency:** Which surgical procedures experience the highest cancellation rates, and what are the primary root causes (e.g., equipment bottlenecks, specialist availability, or patient no-shows)?
- **Emergency Department Outcomes:** How do patient outcomes correlate with arrival triage priority levels (Critical vs. Urgent vs. Non-urgent)? What percentage of critical emergencies result in inpatient admissions?
- **OPD Clinic Effectiveness:** What is the follow-up compliance rate for patients flagged as requiring subsequent clinical evaluations during their initial OPD outpatient visits?

7. DATA INTEGRITY & VALIDATION SAFEGUARDS

Handling four distinct clinical pipelines (OPD, ER, Inpatient, and Surgeries) requires careful synchronization to avoid common relational database generation errors. To prevent silent logic conflicts, we will enforce three core data validation guidelines:

- **Sequential Dependency Flow:** Parent entities (Patients, Doctors, Wards/Beds) are programmatically generated and locked first. Downstream transaction tables sample directly from these validated records using `pandas .sample()`, eliminating orphaned foreign key errors.
- **Chronological Rules:** Every datetime calculation relies on relative math rather than independent random selection. For instance, a patient's admission timestamp must precede their surgery date, which must precede their discharge timestamp.
- **Triage and Flow Outcome Alignment:** For Emergency records, the outcome pathways (e.g., "Admitted") are verified against actual Admissions table foreign keys to maintain logical consistency across clinical tables.

8. CONCLUSION

With guidance from your internship supervisor, expanding the project to encompass the ****11-table schema**** provides an extremely thorough, professional, and realistic operational intelligence database.

By modeling the real-world operational realities of **Indus Hospital**, the database shifts from a simple clinical tracking tool into a robust resource-planning platform. This project serves as an exceptional data engineering showcase, illustrating how relational database design, Python automation, and advanced SQL analysis solve core hospital administrative challenges.

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